

# 1 $H^{(0)}(q) \rightarrow h(k_1)Z(k_2)\gamma(k_3)$ in 2HDM

$$\mathcal{A} = \mathcal{A}^{\mu\nu} \epsilon_{Z\mu}(k_2) \epsilon_{\gamma\nu}(k_3), \quad (1)$$

$$\mathcal{A}^{\mu\nu} = g^{\mu\nu} A_{00} + \sum_{i,j=1}^3 A_{ij} k_i^\mu k_j^\nu. \quad (2)$$

Transversed conditions and kinematics,

$$k_2 \cdot \epsilon_Z(k_2) = 0, \quad k_3 \cdot \epsilon_\gamma(k_3) = 0, \quad (3)$$

$$k_3^2 = 0, \quad k_2^2 = m_Z^2, \quad k_1^2 = m_h^2,$$

$$q_{12} = (k_1 + k_2)^2, \quad q_{13} = (k_1 + k_3)^2, \quad q_{23} = (k_2 + k_3)^2,$$

$$k_3 \cdot k_1 = \frac{(q_{13} - m_h^2)}{2}, \quad k_3 \cdot k_2 = \frac{q_{23} - m_Z^2}{2},$$

$$q_{12} + q_{13} + q_{23} = m_h^2 + m_H^2 + m_Z^2. \quad (4)$$

The photon Ward-identity,

$$k_3^\nu \mathcal{A}_{\mu\nu} = 0. \quad (5)$$

Applying transversed conditions to the amplitude  $\mathcal{A}$ ,

$$\mathcal{A} = A_{00} [\epsilon_Z \cdot \epsilon_\gamma] + \sum_{i,j=1}^3 A_{ij} (k_i \cdot \epsilon_Z(k_2)) (k_j \cdot \epsilon_\gamma(k_3)). \quad (6)$$

The nonzero contributions are those with indices  $i \neq 2$  and  $j \neq 3$ ,

$$\mathcal{A}^{\mu\nu} = A_{00} g^{\mu\nu} + \sum_{\substack{i,j=1 \\ i \neq 2 \\ j \neq 3}}^3 \mathcal{A}_{ij} k_i^\mu k_j^\nu \quad (7)$$

Photon Ward-identity leads to

$$\begin{aligned} 0 &= k_3^\nu \mathcal{A}_{\mu\nu} = A_{00} k_3^\mu + \sum_{\substack{i,j=1 \\ i \neq 2 \\ j \neq 3}}^3 \mathcal{A}_{ij} k_i^\mu (k_3 \cdot k_j) \\ &= k_3^\mu \left( A_{00} + \sum_{\substack{j=1 \\ j \neq 3}} \mathcal{A}_{3j} (k_3 \cdot k_j) \right) + \sum_{\substack{i,j=1 \\ i \neq 2,3 \\ j \neq 3}} \mathcal{A}_{ij} k_i^\mu (k_3 \cdot k_j) \\ &= k_3^\mu \left( A_{00} + \sum_{\substack{j=1 \\ j \neq 3}} \mathcal{A}_{3j} (k_3 \cdot k_j) \right) + k_1^\mu \left( \sum_{\substack{j=1 \\ j \neq 3}} \mathcal{A}_{1j} (k_3 \cdot k_j) \right) \end{aligned} \quad (8)$$

$k_i^\nu$  and  $q^\mu$  are constrained by 4-momentum conservation at each vertices, thus the two,  $k_3^\mu$  and  $k_1^\mu$  out of three  $k_i^\mu$  are free four-vectors. Hence,

$$\left( \mathcal{A}_{00} + \sum_{\substack{j=1 \\ j \neq 3}} \mathcal{A}_{3j}(k_3 \cdot k_j) \right) = 0, \quad \sum_{\substack{j=1 \\ j \neq 3}}^3 A_{1j}(k_3 \cdot k_j) = 0. \quad (9)$$

then,

$$\mathcal{A}_{00} = -\mathcal{A}_{31}(k_3 \cdot k_1) - \mathcal{A}_{32}(k_3 \cdot k_2), \quad \mathcal{A}_{12}(k_3 \cdot k_2) = -\mathcal{A}_{11}(k_3 \cdot k_1). \quad (10)$$

7 now reads

$$\begin{aligned} \mathcal{A}^{\mu\nu} &= \mathcal{A}_{00}g^{\mu\nu} + \mathcal{A}_{11}k_1^\mu k_1^\nu + \mathcal{A}_{12}k_1^\mu k_2^\nu + \mathcal{A}_{31}k_3^\mu k_1^\nu + \mathcal{A}_{32}k_3^\mu k_2^\nu \\ &= \mathcal{A}_{31}(-(k_3 \cdot k_1)g^{\mu\nu} + k_3^\mu k_1^\nu) + \mathcal{A}_{32}(-(k_3 \cdot k_2)g^{\mu\nu} + k_3^\mu k_2^\nu) + \mathcal{A}_{11} \left( -\frac{k_3 \cdot k_1}{k_3 \cdot k_2} k_1^\mu k_2^\nu + k_1^\mu k_1^\nu \right) \\ &= \mathcal{A}_{31}P_1^{\mu\nu} + \mathcal{A}_{32}P_2^{\mu\nu} + \mathcal{A}_{11}P_3^{\mu\nu} \end{aligned} \quad (11)$$

The total decay width,

$$\Gamma_{H \rightarrow hZ\gamma} = \frac{1}{256\pi^3 m_H^3} \int_{q_{23}^{\min}}^{q_{23}^{\max}} dq_{23} \int_{q_{13}^{\min}}^{q_{13}^{\max}} dq_{13} \sum_{\text{pol}} |\mathcal{A}|^2 \quad (12)$$

where

$$\begin{aligned} \sum_{\text{pol}} |\mathcal{A}|^2 &= \mathcal{A}_{\mu\alpha} \mathcal{A}_{\nu\beta} \mathcal{P}_Z^{\mu\nu} \mathcal{P}_\gamma^{\alpha\beta} \\ &= |F_1|^2 \chi_1 + |F_2|^2 \chi_2 + |F_3|^2 \chi_3 + F_1 F_2 \chi_4 + F_1 F_3 \chi_5 + F_2 F_3 \chi_6 \end{aligned} \quad (13)$$

with

$$\begin{aligned} \mathcal{P}_\gamma^{\alpha\beta} &= \sum_{\lambda_Z=0,\pm 1} \epsilon_\gamma^{(\lambda)\alpha} \epsilon_\gamma^{(\lambda)\beta} = -g^{\alpha\beta} + \dots \\ \mathcal{P}_Z^{\mu\nu} &= \sum_{\lambda_Z=0,\pm 1} \epsilon_Z^{(\lambda)\mu} \epsilon_Z^{(\lambda)\nu} = -g^{\mu\nu} + \frac{k_2^\mu k_2^\nu}{m_Z^2} \end{aligned} \quad (14)$$

and  $A_{31} = F_1, A_{32} = F_2, A_{11} = F_3$ ,

$$\begin{aligned}
\chi_1 &= \frac{(m_h^2 - q_{13})(m_h^2 q_{23} - m_Z^2 + m_Z^2(q_{12} - q_{13} + q_{23}) - q_{12}q_{23})}{4m_Z^2} + \frac{1}{4}(D-4)(m_h^2 - q_{13})^2, \\
\chi_2 &= -\frac{1}{4}(m_Z^2 - q_{23})^2 + \frac{1}{4}(D-4)(m_Z^2 - q_{23})^2, \\
\chi_3 &= \frac{(m_h^4 - 2m_h^2(m_Z^2 + q_{12}) + (m_Z^2 - q_{12})^2)}{4(m_Z^2 - m_h^2 q_{23})^2} \{m_h^4 q_{23} + m_h^2(m_Z^2(q_{12} - q_{13} - q_{23}) \\
&\quad + q_{23}(-q_{12} - q_{13} + q_{23})) + q_{13}(m_h^4 - m_Z^2(q_{12} - q_{13} + q_{23}) + q_{12}q_{23})\}, \\
\chi_4 &= \frac{3}{2}(m_h^2 - q_{13})(m_Z^2 - q_{23}^2) + \frac{1}{4}(D-4)(m_h^2 - q_{13})(m_Z^2 - q_{23}), \\
\chi_5 &= \frac{(m_h^2 - q_{13})(m_h^4 - 2m_h^2(m_Z^2 + q_{12}) + (m_Z^2 - q_{12})^2)}{4m_Z^2}, \\
\chi_6 &= \frac{(m_h m_Z(m_h + m_Z) - m_h q_{23} - m_Z q_{13})(m_h q_{23} - m_Z(m_h(m_Z - m_h) + q_{13}))}{m_Z^2 - q_{23}}.
\end{aligned} \tag{15}$$

## 2 Grouped Diagrams

There is no vertex  $HhZ\gamma$  in 2HDM at tree-level, this process is loop-induced. The relevant contribution for the vertex correction for the process in  $R_\xi$  gauge (t'Hooft gauge  $\xi = 1$ ),

$$\mathcal{A} = \mathcal{A}_f + \mathcal{A}_{W,G,u^\pm} + \mathcal{A}_{H^\pm} + \mathcal{A}_{H,G,W,u^\pm} \tag{16}$$

The notation  $A_{\text{particles}}$  indicates the internal loop fields contributing to the process. There is also the higgstrahlung contribution that radiative off  $h$  final state in one leg and have virtual reduced diagrams (blob) to  $\phi \rightarrow Z\gamma$  where  $\phi = h, H$  in another.

Diagrams are grouped to its Gauge dependences. Higgs and fermion contributions are Gauge independent, while Gauge-boson, Goldstone and Ghost contribution with mixed Higgs contributions are Gauge dependence. The form factors  $F_i$  can be factored further by

$$F_i = F_i^{\text{Box}} + F_i^{\text{Triangle}} + F_i^{\text{Self}} \tag{17}$$

## 3 Higgs loop